Xiaomeng Yang

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Research focus: Video generation and world models; efficient diffusion models; multimodal learning and visual text understanding.

Education

Northeastern University Sep. 2024 – Present
Ph.D. in Computer Engineering; expected April 2029 Boston, MA
Advisors: Yanzhi Wang and Xuan Zhang

University of Chinese Academy of Sciences Sep. 2021 – June 2024
M.S. in Computer Application Technology; advisor: Yu Zhou Beijing, China

University of Illinois Urbana-Champaign & Zhejiang University Sep. 2017 – June 2021
Dual-degree program: B.S. in Computer Engineering with Honors (UIUC);
B.E. in Electronic and Computer Engineering (ZJU)

Research & Industry Experience

Event-Aligned Visual Action Reasoning June 2026 – Present
Northeastern University | Research Assistant; advisor: Yanzhi Wang, Xuan Zhang Boston, MA

- Developed an event-aligned visual action reasoning framework that learns adaptive-rate video imagination, aligning visual-action chunks with task-relevant interactions.
- Preserved fine-grained visual evidence around interactions and jointly learned execution validity to skip repeated action groups while retaining a fixed-length model output.
- Improved success on DOMINO dynamic manipulation tasks from 32.57% to 42.83% over LingBot-VA; evaluated transfer across motion levels and performance on RoboTwin 2.0 tasks.

Snap Inc. Sep. 2025 – May 2026
Research Intern, Creative Vision Santa Monica, CA

- Developed Prompt2Effect (ECCV 2026), a hypernetwork that generates effect-specific LoRA weights in one forward pass for image-to-video model specialization.
- Introduced SVD canonicalization and base-weight conditioning to resolve LoRA factorization ambiguity and stabilize high-dimensional weight synthesis.
- Reduced per-effect adaptation from 56 GPU-hours of training to 3.3 seconds of LoRA synthesis; generated weights also enabled approximately 10× faster downstream optimization.

Efficient Diffusion Generation Nov. 2024 – May 2025
Northeastern University | Research Assistant; advisor: Yanzhi Wang Boston, MA

- Developed ALTER (Neurips 2025), jointly learning layer pruning and timestep-conditioned expert routing to specialize sparse diffusion substructures across denoising timesteps.
- Achieved 3.64× inference speedup with 35% sparsity and 20 denoising steps, compared with a 50-step Stable Diffusion v2.1 baseline.

Scene Text Recognition & Visual Question Answering Oct. 2020 – June 2024
UCAS | advisor: Yu Zhou; Tomorrow Advancing Life | Research Intern, Nov. 2022 – June 2024 Beijing, China

- Developed diffusion-based parallel decoding and masked/permuted context learning for scene text recognition, leading to first-author papers in IJCV 2025 and IEEE SPL 2024.
- Co-developed TextVQA models that couple text reading with visual reasoning, using VQA supervision to optimize text reading and improve robustness to OCR errors (ACM MM 2021; Pattern Recognition 2023).

Selected Publications

Name in bold; * denotes equal contribution.

Prompt2Effect: Training-Free Image-to-Video Model Specialization via LoRA Generation
Xiaomeng Yang, Yanyu Li, Gordon Guocheng Qian, Ivan Skorokhodov, Viacheslav Ivanov, Avalon Vinella, Xuan Zhang, Yanzhi Wang, Sergey Tulyakov, Anil Kag.
European Conference on Computer Vision (ECCV), 2026.

ALTER: All-in-One Layer Pruning and Temporal Expert Routing for Efficient Diffusion Generation
Xiaomeng Yang*, Lei Lu*, Qihui Fan, Changdi Yang, Juyi Lin, Yanzhi Wang, Xuan Zhang, Shangqian Gao.
Advances in Neural Information Processing Systems (NeurIPS), 2025.

IPAD: Iterative, Parallel, and Diffusion-Based Network for Scene Text Recognition
Xiaomeng Yang, Zhi Qiao, Yu Zhou.
International Journal of Computer Vision (IJCV), 2025.

Linguistics-aware Masked Image Modeling for Self-supervised Scene Text Recognition
Yifei Zhang, Chang Liu, Jin Wei, **Xiaomeng Yang**, Yu Zhou, Can Ma, Xiangyang Ji.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.

Zerosim: Zero-Shot Analog Circuit Evaluation with Unified Transformer Embeddings
Xiaomeng Yang, Jian Gao, Yanzhi Wang, Xuan Zhang.
IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2025.

Masked and Permuted Implicit Context Learning for Scene Text Recognition
Xiaomeng Yang, Zhi Qiao, Jin Wei, Dongbao Yang, Yu Zhou.
IEEE Signal Processing Letters, 2024.

Accurate and Robust Scene Text Recognition via Adversarial Training
Xiaomeng Yang, Dongbao Yang, Zhi Qiao, Yu Zhou.
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2024.

Beyond OCR+VQA: Towards End-to-End Reading and Reasoning for Robust and Accurate TextVQA
Gangyan Zeng, Yuan Zhang, Yu Zhou, **Xiaomeng Yang**, Ning Jiang, Guoqing Zhao, Weiping Wang, Xu-Cheng Yin.
Pattern Recognition, 2023.

Beyond OCR+VQA: Involving OCR into the Flow for Robust and Accurate TextVQA
Gangyan Zeng, Yuan Zhang, Yu Zhou, **Xiaomeng Yang**.
ACM International Conference on Multimedia (ACM MM), 2021.

Manuscripts

Event-Aligned Visual Action Reasoning for World Action Models
Xiaomeng Yang, Yushu Wu, Yi Gao, Yuhao Lei, Xuan Zhang, Pu Zhao, Yanzhi Wang.
Under reivew, 2026.

Zerosim+: A Unified Transformer Surrogate for Universal Zero-Shot Analog Circuit Performance Evaluation
Xiaomeng Yang, Jiaqing Li, Jian Gao, Yanzhi Wang, Xuan Zhang.
Under review, 2026.

Additional Research

Transformer-Based Surrogates for Analog Circuit Evaluation Northeastern University
Advisor: Xuan Zhang | Nov. 2024 – Apr. 2025; Jan. 2026 – Sep. 2026

Developed ZeroSim for zero-shot analog circuit performance prediction; extended the framework to multiple circuit families in ZeroSim+.

Technical Skills

Programming: Python, C/C++, Java, JavaScript, HTML/CSS.
Frameworks & tools: PyTorch, TensorFlow, NumPy, SciPy, Linux, Git.
Languages: Chinese (native), English (advanced).